Akshay Pradeep Hiranandani	07557700	Director
Simrun Mehta	09118938	Director
Pratik Pravin Agarwal	03040062	Director
Ravi Thanvi	09631208	Additional Director

Source: MCA

Brief profile of the above Board of Directors is given below:

(a) Akshay Pradeep Hiranandani - At Serentica, Mr. Akshay heads the Corporate Finance division, overseeing capital raising and allocation strategies for the company's sustainable energy platform, which is committed to delivering round-the-clock green power to some of India's largest industrial consumers. With nearly two decades of experience — including over fifteen years in Corporate Finance and Project Development — he has successfully raised more than USD 1.5 billion through debt and equity for both domestic and international renewable energy developers. He has also led the development and execution of over 1 GW of renewable energy projects. Before joining Serentica, Mr. Akshay led the Structured Investments and Project Finance vertical at Piramal Capital, where he managed, deployed, and exited investments totalling approximately USD 1 billion in infrastructure.

(b) Simrun Mehta – Ms. Simrun Mehta is a Director with KKR's Private Equity team, having joined the firm in 2012. She plays a key role in driving KKR's investments in India, particularly across the consumer, retail, and healthcare sectors. Over the years, she has been actively involved in several of KKR's investments spanning diverse industries.

Ms. Mehta currently serves on the boards of Vini Cosmetics, Lighthouse Learning, Avendus Capital, RE Sustainability, Ness Digital Engineering, and Serentica Renewables. Before joining KKR, she worked with Nomura's fixed income division and earlier with the commodities team at Lehman Brothers. She holds a Bachelor of Arts (Honours) degree in Mathematics from St. Stephen's College.

(c) Pratik Pravin Agarwal – Mr. Pratik Agarwal is the Chairman of Serentica Renewables, a green energy company dedicated to delivering round-the-clock, carbon-free power to India's large industrial consumers. He also serves as the Managing Director of Sterlite Power and is a Non-Executive Director at Sterlite Technologies Limited and Sterlite Investment Managers Ltd. Previously, he has held directorial positions at Sterlite Ports, Vizag General Cargo Berth Ltd, and Speedon Networks Limited, while also leading Vedanta Group's investor relations and strategic initiatives. Vedanta Limited is a diversified natural resources company with annual revenues of approximately USD 17 billion. He holds a Bachelor of Science degree from the Wharton School, University of Pennsylvania, and an MBA from the London Business School.

(d) Ravi Thanvi – Mr. Ravi Thanvi is a Principal with KKR's Asia-Pacific Infrastructure team, based in Mumbai, where he oversees the firm's energy investments in India. He brings over 13 years of experience in private equity and investment banking, specializing in the energy and infrastructure sectors across India and Southeast Asia. Before joining KKR, Mr. Ravi spent six years at Actis, where he played a pivotal role in expanding the firm's energy and infrastructure portfolio in India and Southeast Asia. During his tenure, he helped deploy over USD 1 billion across projects in renewables, gas, and toll roads. Earlier in his career, he worked in investment banking at Goldman Sachs and HSBC, advising global clients on mergers and acquisitions as well as strategic fundraising related to investments in India. Mr. Ravi holds a Master's degree in Mathematics from the Indian Institute of Technology, Kharagpur (IIT-KGP) and an MBA from the Indian Institute of Management, Ahmedabad (IIM-A).

3.5.2 Past Financials of SRIPL

Past Financials of SRIPL is given below:

Particulars	**Rs. in crore unless otherwise specified**	
	FY25	**FY24**
Revenue from Operations	797.14	142.55
Other Income	78.09	18.35
Total Income	875.23	160.90
Operating EBIDTA	93.59	55.95
Total EBIDTA	171.68	74.30
Operating EBIDTA %age	11.74%	39.25%
Total EBIDTA %age	19.62%	46.18%
Profit Before Tax (PBT)	78.89	37.98
Profit After Tax (PAT)	85.36	27.21
PBT %age	9.01%	23.60%
PAT %age	9.75%	16.91%
Total Equity	801.92	666.21
Total Outside Liabilities (TOL)	2344.26	756.36
Tangible Net Worth (TNW)	801.92	665.07
TOL/ TNW	2.92	1.14
Current Assets	1753.95	392.40
Current Liabilities	1769.29	280.47
Current Ratio	0.99	1.40

Source: Audited Balance Sheet

Our Comments:

- ***Revenue from Operations:*** *SRIPL is in the business of build, own, manage, operate and maintain hydel, solar, wind, geothermal and other renewable energy power projects. SRIPL primarily earns its Revenue from Operations by providing EPC Services and Management Fee Income. During FY25, revenue from operations increased multifold to Rs.797.14 crore from Rs.142.55 crore in FY24. The increase is mainly due to revenue booked from EPC Services amounting to Rs.637.28 crore.*
- ***Other Income:*** *Other Income earned during FY25 was Rs.78.09 crore. This primarily constitutes Interests on Bank Deposits amounting to Rs.45.47 crore and Interests on Loan given to Related Parties amounting to Rs.31.99 crore.*
- ***EBIDTA:*** *Even though Revenue from Operations of SRIPL increased from Rs.142.55 crore in FY24 to Rs.797.14 crore in FY25, EBIDTA levels dropped from being at 46.18% in FY24 to 19.62% in FY25. Primary reason attributable to drop in EBIDTA levels is increase in Employee Benefit Expenses (from Rs.33.41 crore in FY24 to Rs.67.88 crore in FY25) and increase in Other Expenses (from Rs.53.19 crore in FY24 to Rs.81.51 crore in FY25).*
- ***Total Equity:*** *Total Equity of SRIPL increased from Rs.666.21 crore to Rs.801.92 crore backed by ploughing back of profits during the year.*
- ***Total Outside Liabilities (TOL):*** *TOL of SRIPL increased from Rs.756.36 crore to Rs.2344.26 crore. This can primarily be attributed towards increase in Other Current Liabilities rising to Rs.1267.90 crore in FY25 from Rs.5.77 crore in FY24. Other Current Liabilities primarily constitutes of Rs.1239.05 crore being Advance from Customers.*
- ***TOL/ TNW:*** *TOL/ TNW of SRIPL has increased from 1.14 times in FY24 to 2.92 times in FY25 primarily due to increase in Other Current Liabilities.*
- ***Current Ratio:*** *Current Ratio of SRIPL dropped to 0.90 times in FY25 from being at 1.40 times in FY24. This is majorly due to increase in Short Term Borrowings and Other Current Liabilities.*
- ***Contingent Liabilities:*** *SRIPL does not have any Contingent Liabilities as on 31st March'2025.*

4 RBI GUIDELINE

4.1 RBI Guideline

For the applicability of RBI Circular and Norms related to the proposed project finance, we have referred following circulars:

(i) Master Direction – Reserve Bank of India (Transfer of Loan Exposures) Directions, 2021 (RBI/DOR/2021-22/86 - DOR.STR.REC.51/21.04.048/2021-22 - dated 24th September 2021, updated on 28th December 2023).

Also, due to change in project cost and loan amount, following RBI notification has been referred:

(ii) Notification - RBI/2025-26/59 - DOR.STR.REC.34/21.04.048/2025-26 -dated 19th June 2025.

Compliance status of Master Direction – Reserve Bank of India (Transfer of Loan Exposures) Directions, 2021 dated 24th September 2021, updated on 28th December 2023:

Table 17: Compliance with Master Direction of RBI (Transfer of Loan Exposures)

<table>
<tr><th>Circular</th><th>Compliance</th></tr>
<tr><td>Clause No 11: Loan transfers should result in transfer of economic interest without being accompanied by any change in underlying terms and conditions of the loan contract usually. In all cases, if there are any modifications to terms and conditions of the loan contract during and after transfer (eg. in take-out financing), the same shall be evaluated against the definition of ‘restructuring’ provided in Paragraph 1 of the Annex to the Reserve Bank of India (Prudential Framework for Resolution of Stressed Assets) Directions 2019, dated June 7, 2019.</td><td>As we understand the following change has been proposed in loan terms:
<table>
<tr><th>Particulars</th><th>Change Proposed</th><th>Remarks</th></tr>
<tr><td>Loan Amount</td><td>Being increased From Rs 2971.27 Cr to Rs 3077.21, i.e., additional loan of Rs 106 Cr.</td><td>Since the loan amount proposed to be modified, Clause no 11 will be applicable

Complied (refer note below)</td></tr>
<tr><td>ROI</td><td>From 9.45% to 8.75% (pre-COD)</td><td>This is Commercial decision, not to be considered</td></tr>
</table></td></tr>
</table>

<table>
<tr><th>Circular</th><th>Compliance</th></tr>
<tr><td></td><td>
<table>
<tr><td></td><td>From 9.20% to 8.25% (post COD)</td><td rowspan="2">as change in terms and condition hence not applicable.</td></tr>
<tr><td>Pre-Payment Penalty</td><td>Reduced from 2% to 1%</td></tr>
</table>
Note:
As per Paragraph 1 of Annex 1 to the Reserve Bank of India (Prudential Framework for Resolution of Stressed Assets) Directions 2019, dated June 7, 2019, Restructuring is an act in which a lender, for economic or legal reasons relating to the borrower's financial difficulty, grants concessions to the borrower.

Since the project is under implementation, the critical financial difficulty shall be infusion of promoters' contribution for the project. We have reviewed the credit rating of the parent company of the borrower and found the latest ICRA Rating as [ICRA] AA- (stable) dated 19th November 2025. Thus, we understand, the borrower / parent does not have financial difficulty; however, Lender may verify the CIBIL & other necessary records of borrower to verify borrower's creditworthiness.
</td></tr>
<tr><td>Clause No 10: General Conditions applicable for all loan transfers under Reserve Bank of India (Transfer of Loan Exposures) Directions, 2021 (RBI/DOR/2021-22/86 - DOR.STR.REC.51/21.04.048/2021-22 -dated 24th September 2021, updated on 28th December 2023)</td><td>The lenders must put in place a comprehensive Board approved policy for transfer and acquisition of loan exposures under these guidelines.

Compliance: Lender to note and do the needful.</td></tr>
</table>

Circular	Compliance
Minimum Holding Period	Minimum Holding Period (MHP) has been defined in Chapter III as mentioned in the Circular detailed above. This provisions is Not Applicable for the present proposal as the of *Transfer of Loan Accounts of Borrower by a Lender to Other Lenders, at the request/ instance of the Borrower.*

RBI, Project Finance Directions, 2025 dated 19th June 2025 which is effective from 01st October 2025:

Table 18: Compliance with Project Finance Direction of RBI

Clause	Basis of Applicability
As per Clause 7 of the Circular, "The RBI Circular dated 19th June 2025 shall not apply to projects where financial closure has been achieved as on the effective date. Such projects shall continue to be guided by the existing prudential guidelines on project finance, which otherwise shall be treated as repealed. However, any resolution of a fresh credit event and/or change in material terms and conditions in the loan contract in such projects, subsequent to the effective date, shall be as per the guidelines contained in these Directions".	As we understand, the project has already achieved financial closure, however, since the loan is proposed to be taken over with enhanced debt amount and thus may be treated as fresh credit event and hence the Circular shall be applicable.

Compliance Status

Table 19: Compliance Status

Circular	Compliance
Deferment of DCCO (Clause (a) of Para 26)	DCCO remains same as per the previous sanction and thus not applicable.
Financing of Cost Overruns for Projects under Implementation (Clause (b) of Para 26)	A lender may finance, as part of a resolution plan, cost overrun and classify the account as 'Standard', under following conditions: Funding of Cost overrun is allowed under following conditions: **Clause** | **Compliance**

Circular	Compliance	
	i. Cost overrun up to a maximum of 10% of the original project cost, in addition to IDC.	Excluding IDC cost overrun is only 3.75% and hence complied. (refer note -1 below the table for detailed calculation)
	ii. Cost overrun is financed through SBCF specifically sanctioned by the lender at the time of financial closure;	• The instant project is considered as Infrastructure project. • Hence, additional funding without SBFC can also be done, however, such additional funding shall be priced at a premium to what would have been applicable on a pre-sanctioned SBCF. May be treated as complied provided lender charging premium on price of the additional loan basis their internal risk assessment.
	iii. For infrastructure projects, in cases where SBCF was not sanctioned at the time of financial closure, or was sanctioned but not renewed subsequently, such additional funding shall be priced at a premium to what would have been applicable on a pre-sanctioned SBCF. Lenders shall ensure that the loan-contracts ab-initio specify the additional risk premium to be charged on such SBCF, which may be revised upwards based on actual risk assessment at the time of sanction of such facilities.	
	iv. The financial parameters like D/E ratio, external credit rating (if any) etc. remain unchanged or are enhanced in favour	• No change in debt equity ratio hence complied. • As on date, the Company does not

Circular	Compliance
	of the lender post such cost overrun funding | have any external credit rating. We understand the new lender shall stipulate fresh credit rating within 6 months of 1st disbursement. Since there is no external rating, this point is considered to be not applicable.
As per Clause no 33, for accounts which have availed DCCO deferment as per Part K of these Directions and are classified as 'standard', lenders shall maintain additional specific provisions of 0.375% for infrastructure project loans and 0.5625% for non-infrastructure project loans (including CRE and CRE-RH), for each quarter of deferment, over and above the applicable standard asset.	The DCCO deferment has been approved by previous lender within the embit of Master Circular of April 2025, i.e., prior to the present circular coming into force (from 01st October 2025) and hence not applicable.

Table 20: Calculation of Cost Overrun excluding IDC

Particulars	REC Cost basis reduced capacity (840 MWp)	Revised Cost for 840 Mw
Land	41.40	70.00
PV Module	2034.24	3141.23
BOP	1106.99	
Overheads	33.28	

Particulars	REC Cost basis reduced capacity (840 MWp)	Revised Cost for 840 Mw
Total Hard Cost	**3215.91**	**3211.23**
Preoperative Cost	36.00	109.28
Financing Cost	17.90	94.04
Contingencies	66.56	64.22
Working capital margin money	11.79	11.79
DSRA	93.62	80.32
IDC	127.01	275.63
Total Soft Cost	**352.88**	**635.28**
Total Project Cost	**3568.79**	**3846.51**
Project Cost Excluding IDC	3441.78	3570.88
Increase in Project Cost excluding IDC	3.75%	

5 PROJECT DESCRIPTION

5.1 Project Description

Vedanta Group a multi-national conglomerate operating in various sectors depends heavily on 24x7 power availability to ensure smooth business operations. The group in order to have energy security along with availability of clean and green power and to increase its foothold in the renewable energy sector have incorporated ***Serentica Renewables India 9 Private Limited*** for setting up a Solar Power Generating Project having capacity of 840 MWdc/ 600 MWac located at ***Kotha & Harba Villages, Tehsil – Fatehgarh, District – Jaisalmer, Rajasthan***.

5.2 Project Snapshot

A brief snapshot of the proposed project is provided as under:

Table 21: Project Snapshot

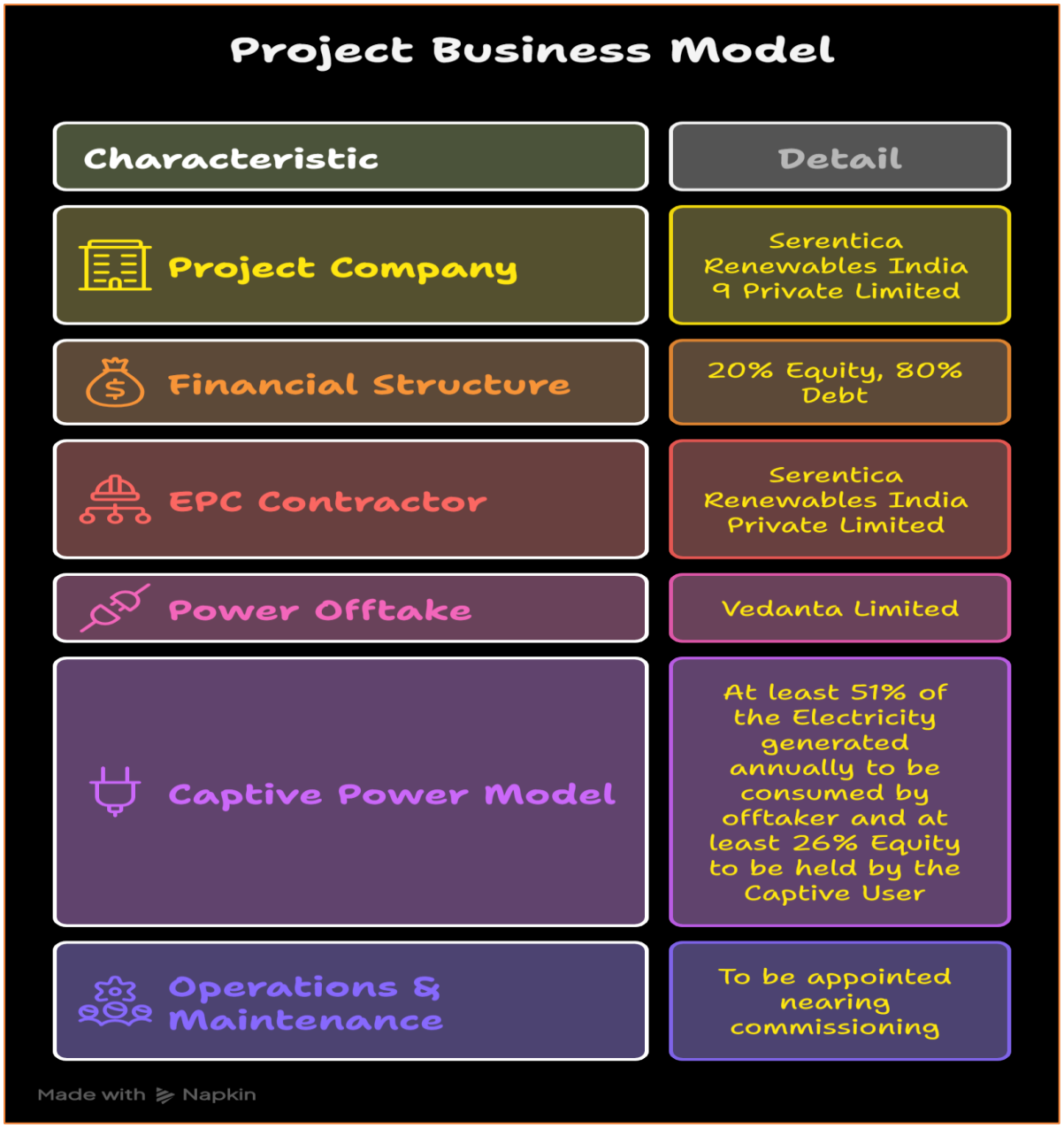
<table>
<tr><th>Particulars</th><th>Details</th></tr>
<tr><td>Project</td><td>Setting up a Solar Power Project having Capacity of 840 MWdc/ 600 MWac located at Kotha & Harba Villages, Tehsil – Fatehgarh, District – Jaisalmer, Rajasthan</td></tr>
<tr><td>Location of the Project</td><td>Location of the project is described as under:
<table>
<tr><th>Particulars</th><th>Details</th></tr>
<tr><td>Villages</td><td>Kotha & Harba</td></tr>
<tr><td>Tehsil</td><td>Fatehgarh</td></tr>
<tr><td>District</td><td>Jaisalmer</td></tr>
<tr><td>State</td><td>Rajasthan</td></tr>
<tr><td>Latitude</td><td>26.40° N</td></tr>
<tr><td>Longitude</td><td>70.96° E</td></tr>
</table></td></tr>
<tr><td>Land</td><td>The project is being set up on Lease Hold Land admeasuring ~2300 acres.

At present the company has in its possession 2138 acres of land, balance portion is expected to be acquired by December’2025.</td></tr>
<tr><td>Project Capacity</td><td>The present proposal entails setting up the following capacity:
<table>
<tr><th>Particulars</th><th>MWdc</th><th>MWac</th><th>DC to AC Ratio</th><th>PLF at P90</th></tr>
<tr><td>Project Capacity</td><td>840</td><td>600</td><td>1.40</td><td>20.33% (at DC level) and 28.46% (at AC level)</td></tr>
</table></td></tr>
</table>

Particulars	Details
Power Evacuation	Power generated from the project shall be evacuated at 400 kV Evacuation voltage level at CTU of Fatehgarh Rajasthan. The total connectivity maximum capacity as approved by CTU is 900 MW. 400/33kV plant switchyard shall have two transformer bay, two main bus, one bus coupler and one line bay for the 400kV S/C TL to connect to CTU (approx. 12.50 km)
Transmission Line	The project involves construction of 400kV having 36 towers with a Stringing Length of 12.50 kms. As per project progress as on 31st October'2025, out of the total 36 towers, Foundation of 34 towers has been completed, also ROW for 34 out of 36 towers is also in possession. Further, Erection of 18 towers is completed. Stringing of 0.05 km out of 12.50 km is completed.
Major Equipment Specifications	The major equipment specification forming part of the Solar Power Plant is provided as under: **Particulars** / **Details** Solar PV Modules: 690 Wp N-Type TOPcon Bifacial supplied by Waaree Energies Limited Inverter: 3300 kw Central Inverters supplied by Sineng Electric Co. Limited
Power Offtake	Power generated from the Solar Plant will be sold to ***Vedanta Limited (VL)***. A Power Delivery Agreement (PDA) has been executed between SRI9PL and VL for offtake of 600 MWac of Contracted Capacity under Captive Consumption model at a fixed tariff of $ Cents 3.05 per unit for a period of 25 years from the COD of the entire Contracted Capacity
SCOD	***30th June'2026***

5.3 Project Business Model

The company proposes to develop the Solar Power Plant under ***"captive model"*** wherein it will enter into a Power Purchase Agreement (PPA) with a private customer (Offtaker) under group captive policy which will hold at least 26% stake in the company. An exhibit of Project Business Model is provided hereunder:



5.4 Site Location

The project is coming up at Kotha & Harba Villages, Tehsil – Fatehgarh, District – Jaisalmer, Rajasthan. The site is located ~75 km from the Jaisalmer Airport. The site is located at Latitude 26.40° N and Longitude 70.96° E with an Elevation of 226 meters above Mean Sea Level (MSL).

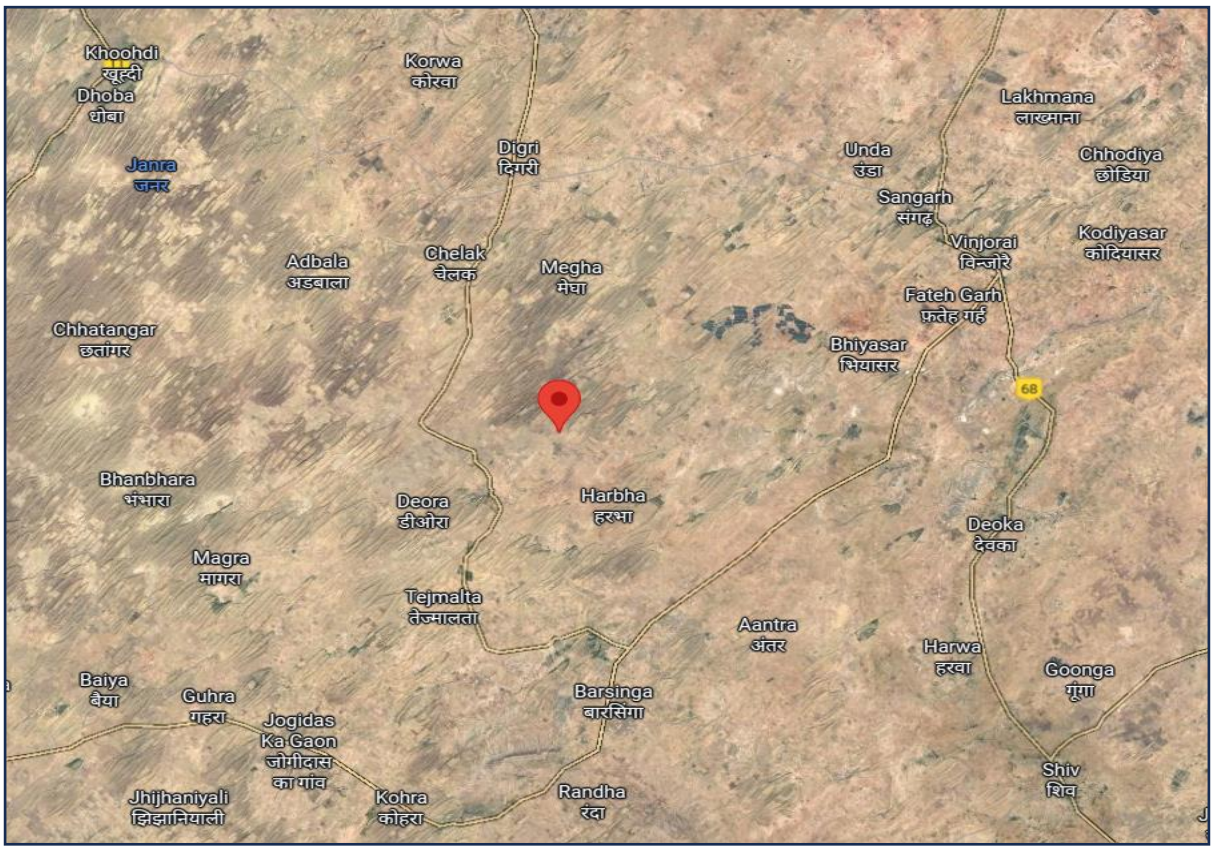


Exhibit: Project Location

The project is located in Jaisalmer district of Rajasthan which is the western region of the state. The climate in this region is Hot Arid with extremely hot temperatures in from April through July. A detail of geographical advantage of setting up Solar Power Project in Jaisalmer is given below:

Table 22: Geographical Advantage of Setting up Solar Project in Jaisalmer

Geographical Advantage	Detail
Terrain and Land	Vast, largely barren desert tracts with low competing land-use → easier aggregation of large, contiguous parcels; simpler site preparation (minimal cut-fill) Gentle topography (mostly flat/sandy plains) → lower civil costs for piling, cable trenches, internal roads, and O&M access

Geographical Advantage	Detail
Climate	Arid zone with very low annual rainfall → fewer washout/erosion events and minimal flooding risk Inland location, negligible cyclone risk; low seismicity (Zone II) → reduced extreme-weather/earthquake exposure
Solar Resource	Very high insolation: ~5.5–6.5 kWh/m^2/day GHI; >300 sunny days/year → strong CUF potential for both fixed-tilt and single-axis trackers Low cloud cover and minimal seasonal variability compared to coastal/monsoon-heavy regions → more predictable generation
Grid & Evacuation	Proximity to Rajasthan's Green Energy Corridor nodes and large pooling substations around Fatehgarh/Jaisalmer; availability of 220/400/765 kV bays and ISTS connectivity in the broader cluster → scalable, multi-GW evacuation

5.4.1 Site Connectivity

The site is well connected through State Highway 21 which is located at an ~5 km distance from the project location. Details regarding site connectivity is given below:

Table 23: Site Connectivity

Nearest State Highway	SH 21 - ~5 km
Nearest National Highway	NH 68 - ~25 km
Nearest Railway Station	Jaisalmer Railway Station - ~75 km
Nearest Airport	Jaisalmer Airport - ~65 km Jodhpur Airport - ~215 km

Our Comments:

- *Overall access to the project site is considered to be good.*
- *The project site lies within 15 km of the existing 400 kV Fatehgarh substation, allowing short transmission corridors and reduced line-loss exposure. Primary access is via the Jaisalmer–Ramgarh highway and a network of village roads capable of accommodating module and transformer transport.*

5.5 Land Details

Solar power plants, while being clean and renewable, are very land intensive. The amount of land needed per kilowatt of installed capacity depends on the technology used and the spacing between solar panels (the pitch), which varies with the location's latitude. Sites must be large enough to accommodate the desired power output without sacrificing energy production due to shading between rows of panels.

The area required per kWp of installed power varies with the technology chosen and the distance between rows of modules (the pitch) required avoiding significant inter-row shading varies with the site latitude. Sites should be chosen with sufficient area to allow the required power to be installed without having to reduce the pitch to levels that cause unacceptable yield loss.

A well designed a well-designed Solar Power Plant requires 2.50 acres to 3.00 acres of land for per MWdc of capacity considering the solar panel array, control rooms, infrastructure, and other facilities.

Basis the site location, technology and other aspects envisaged for the project; the estimated land requirement for the entire project is 2300 acres (including land for Pooling Sub-station).

Table 24: Estimated Land Requirements

Particulars	Area in Acres
Estimated Project Land required for setting up the project	2300
Land already under possession	2138
Balance Land required to be acquired	**162**

Source: Company Information

LSI Comments:

- *Surface Land requirement per acre of Solar Project complies with prevailing industry standards.*
- *Land acquisition is in a mix of leasehold basis (both Private and Government) along with outright purchase.*
- *Almost entire land required for the project is in possession of the Company. As informed by the Company, balance land will be acquired by December'2025.*
- *As per Rajasthan Renewable Energy Policy, 2023, developer shall be permitted to set-up Power Project/ Farm on private agriculture land without the requirement of land conversion in accordance with the provisions of Rajasthan Tenancy Act 1955 and Rajasthan Land Revenue Act 1956 and the rules made there under.*
- *Lender(s) to the project to conduct Land Due Diligence as the same is not part our scope of work.*

5.6 Site Selection and Resource Assessment

5.6.1 Site and Location of Solar

The Solar PV plant is a ground mount solar plant located at Kotha & Harba Villages, Tehsil – Fatehgarh, District: Jaisalmer in the Indian state of Rajasthan. The PV plant lies around the location of 26.40° N and 70.96° E at an altitude of approximately 226 m above mean sea level.

5.6.2 Solar Resource Assessment

LSI has done the detailed resource assessment study based on the documents shared by the company for the review.

The accuracy of solar energy yield predictions largely relies on the precision of the solar resource dataset utilized. Since solar irradiation data is not currently measured at the proposed power plant site, alternative data sources must be employed to estimate the site's irradiation levels.

The solar resource at a location can be characterized by parameters such as global horizontal irradiation (GHI), direct normal irradiation (DNI), and Diffuse Horizontal Irradiation (DHI). These parameters are defined as follows:

- **Global Horizontal Irradiation (GHI):** The global horizontal irradiation is the total solar energy received on a unit area of horizontal surface. It includes energy received from the sun in a direct beam and energy that is received from radiation scattered off the atmosphere arriving from all directions of the sky (diffuse irradiation). The units of GHI are given in kWh/m2. Values are often provided for a period of a day, a month or a year.

- **Diffused Horizontal Irradiation (DHI):** The diffused horizontal irradiation is the energy received from radiation scattered off the atmosphere arriving from all directions of the sky on a unit area of horizontal surface. It is measured in kWh/m2 and values are strongly dependent on weather conditions and the clearness of the air.

- **Direct Normal Irradiation (DNI):** The direct normal irradiation is the total solar energy received on a unit area of surface directly facing the sun. The units of DNI are kWh/m2. The direct normal irradiation is of particular interest for solar installations that track the sun and to concentrating solar technologies as only radiation coming directly from the sun may be focussed by a lens or mirror.

- For modelling of solar PV plants, GHI and DHI are required for calculating the estimated energy yield. In the northern hemisphere, tilting the modules at an angle towards the south increases the total annual global irradiation that is received on the module plane compared to the horizontal plane. This is quantified by the global tilted irradiation. The optimal tilt angle varies primarily with latitude and depends on local weather patterns, ground conditions and plant layout configurations.

- Tilted modules also benefit from irradiation reflected from the ground which is dependent on the ground reflectance, or albedo. Albedo and global tilted irradiation are described below. We have independently optimized the tilt angle and the pitch distance through an iterative process of simulations.

- **Global Tilted Irradiation (GTI):** The global tilted irradiation is the total solar energy received on a unit area of a tilted surface. It includes direct and diffused irradiation along with ground reflected irradiation. The units of GTI are kWh/m2. A transposition model is used for translating horizontal irradiation to tilted irradiation within PV modelling software.

- **Albedo:** The ground albedo or reflectance affects the irradiation on a plane when it is tilted from horizontal and increases the GTI. The albedo is highly site and weather dependent, with typical grass coverings giving an albedo of approximately 0.2 and fresh snow giving an albedo of approximately 0.8, meaning that 20% and 80% respectively of the irradiation is reflected back into the atmosphere.

Table 25: Irradiation Data of the Project Site

Month	**Global Horizontal Insolation (Kwh/m^2)**
January	135.00
February	146.90
March	195.70
April	207.60
May	219.60
June	199.80
July	171.30
August	169.40
September	176.90
October	172.00
November	134.40
December	126.50
Total	**2055.20**

Source: PVsyst report dated 05-09-2025 as provided by the Company

Our Comments:
The typical range of GHI considered good for setting up a Solar PV Power Plant is 1600 to 2200 kWh/m². Basis the PVsyst report, the annual sum of GHI is 2055.20 kWh/m² which is considered adequate for setting up Solar PV Power Plant.

Table 26: Temperature Data of the Project Site

Month	Temperature (°C)
January	16.20
February	21.62
March	26.57
April	32.69
May	33.92
June	33.61
July	32.21
August	29.80
September	29.84
October	29.47
November	24.25
December	17.30
Average	**27.29**

Source: PVsyst report dated 05-09-2025 as provided by the Company

Our Comments:
The typical range of temperature considered good for setting up a Solar PV Power Plant is 25°C to 35°C. Basis the PVsyst report, the yearly average temperature of project site is 27.29°C, which is considered good for setting up Solar PV Power Plant.

Overall Comments on Site Selection and Resource Assessment:
The annual energy yield of a PV plant is significantly influenced by the site's actual solar resource and ambient temperature. For energy estimation purposes, the Solar GIS data source has been utilized.

5.7 Design Review of the Project

The plant's technology philosophy centers on maximizing energy yield while maintaining high system reliability and low maintenance requirements. Each design decision is anchored in bankability, proven field performance, and lifecycle cost optimization.

a) Solar Modules

The project adopts high-efficiency monocrystalline bifacial modules with a nominal power rating of ≈ 690 Wp each, configured for 1,500 V DC operation. Bifacial gain is

expected to enhance annual yield by 6–8 % compared to monofacial designs. Modules are mounted on fixed-tilt structures at 20° angle, optimizing capture for latitude ≈ 26° N. The modules conform to IEC 61215 and IEC 61730 standards, with 25-year performance warranties and ≤ 0.45 % annual degradation assumption.

b) Mounting Structures

Hot-dip-galvanized steel mounting structures are adopted with driven piles embedded to 1.8–2.2 m depth depending on soil resistance. The design accounts for wind speeds up to 180 km/h as per IS 875 (Part III). The layout follows a ground-coverage ratio (GCR) of ≈ 0.45–0.5, balancing bifacial gain and land utilization. Each module table comprises four strings of 29 modules, allowing standardized fabrication and rapid installation.

c) Inverters and Electrical Design

The project uses 188 central inverter units rated ≈ 3.3 MW each, forming the 600 MWac output. Inverters are string-connected to 33 kV transformers through combiner boxes and DC cables sized for ≤ 1.5 % voltage drop. The AC collection system comprises underground XLPE cables feeding into 33 kV switchgear at inverter stations, which aggregate at a 33/400 kV pooling substation. Reactive-power control and harmonic filtering are integrated within the inverter systems to meet grid-code compliance.

d) SCADA and Monitoring

The plant incorporates a centralized SCADA system for real-time data acquisition, remote control, and predictive maintenance. The system interfaces with inverters, transformers, and meteorological sensors through fiber-optic communication loops. Automated alarms for fault detection and performance deviation enable a data-driven O&M approach. The control room at the PSS serves as the operational nerve center.

e) Design Optimization

The design philosophy balances module orientation, inter-row spacing, and DC:AC ratio (≈ 1.40) to maximize yield while ensuring inverter loading is within bankable limits. Electrical losses across DC and AC networks are capped at ≤ 2.5 % overall. Component placement minimizes cable runs and facilitates parallel construction of array blocks. Engineering drawings and construction method statements follow Indian Grid Code and CEA standards.

Basis the above, the technical specifications of the project is given below:

Table 27: Technical Specifications of the Project

Particulars	Details
Project Capacity	840 MWdc/ 600 MWac
PLF (P90 Level)	20.33% (at DC level) and 28.46% (at AC level)

Particulars	Details	
Generation Numbers (P90 Level)	1496.10 GWh	
Average Annual Solar Irradiation	2055.20 kWh/m^2	
Annual Average Temperature	27.29°C	
Tilt Angle	20°	
Solar Modules	OEM	Waaree Energies Limited
	PV Module Type	TOPcon Dual Glass Bifacial
	Unit Size	690 Wp
	No. of Modules per String	29
	No. of Units	12,17,768
	Module Rating	690 Wp at Standard Test Conditions
Inverters	OEM	Sineng Electric Co. Limited
	Inverter Technology	Central Inverters
	Inverter Model	EP-3300-HB-UD
	Unit Size	3300 KW
	No. of Units	188

Our Comments:

- *Waaree Energies Limited (OEM for supply of Solar PV Modules) is one of India's largest Module Manufacturer having manufacturing capacity of 15.10 GW in India and another 1.60 GW in the United States of America.*
- *Sineng Electric Co. Limited (OEM for supply of Inverters) is a global leader in power electronics, specializing in the development and production of Solar PV Inverters, energy storage inverters and Digital Power Products. Some of the clients served by Sineng are ArcelorMittal Nippon Steel India – 606 MW, Raghanesda Solar Park – 200 MW.*
- *OEMs selected for the project are of repute and well established.*

5.8 Off-site Transportation Requirements

We were provided plant location details and based on our site we have observed no such issues in local transportation. Primary access is via the Jaisalmer–Ramgarh highway and a network of village roads capable of accommodating module and transformer transport.

5.9 Energy Yield Assessment

A review of the Energy Yield Assessment (EYA) based on the report as prepared by TÜV Rheinland (India) Private Limited (TÜV) along with summary of the key aspects for the proposed project is provided as under:

Table 28: Solar Energy Assessment

Parameters	Value considered	Our Remarks
Installed Capacity (MWdc)	840 MW	As per EYA Report
Installed Capacity (MWac)	600 MW	As per EYA Report
Pitch and Collector Width (meter)	7.00 * 4.62 & 7.80 * 4.62	As per EYA Report
Global Horizontal Irradiation (kWh/m2)	2055.20	As per EYA Report. Having good irradiations
Near Shading	-2.02%	Mutual shadings of sheds and impacting irradiation. In range of the industry practise
Incidence Angle Modifier (IAM)	-1.75%	IAM are on the lower side and in the range PV syst
Low irradiance	-0.05%	The conversion efficiency of PV modules reduces at low light intensities. The low irradiance loss is calculated within PVsyst for variation in irradiance with respect to the STC conditions. Mono modules are having good conversation rate on low irradiance also (As per Pvsyst)
Module temperature	-7.19%	The characteristics of a PV module are determined at standard temperature conditions of 25˚C. For every °C. temperature rise in temperature; the selected modules reduce in performance. Given the variation in temperature from STC conditions during operation, PV syst has computed loss due to temperature range as 6.5% as per Pvsyst
Soiling	-0.50%	This is the loss due to dust and bird droppings on the PV modules. Depending on the environmental conditions, rainfall frequency and on the plant's O&M module cleaning strategy, soiling loss may vary from1.5% - 2.0% for Manual cleaning in industry practice and 0.8% - 1% for robotic cleaning. But in PVsyst it is considered range 2.0% which is in

Parameters	Value considered	Our Remarks
		range with the industry practices with cleaning cycle of 20-30 per year.
LID	-1.00%	Upon exposure to sunlight, the performance of PV modules decreases over time. Factors affecting the degree of degradation include the quality of materials used in manufacture, the manufacturing process, the quality of assembly and packaging of cells into the modules as well as the O&M regime employed at the site. LID of these modules is good and lower side as it generally 1%. (As per Pvsyst)
Module Mismatch	-1.10%	Mismatch losses are related to the fact that the real modules in string do not all rigorously present the same current/voltage profiles due to statistical variation between them. (As per Pvsyst)
DC Ohmic loss	-1.24%	Average losses are proportionate to average DC voltage drop as per actual DC cables schedule specific to the site. These losses are in range of the Losses considered in the industry of 0.60% (As per Pvsyst)
Inverter Performance	-1.33%	Inverters converts power from DC to AC. Depending on the inverter load; they will not always operate at maximum efficiency. The performance of the inverter was modelled based on. OND file parameters fed into the simulation program. As per the industry practice efficiency loss of the inverter are in range of 1.3%. (As per Pvsyst)
Clipping Losses	0.00%	As per Pvsyst
Plant and Grid Unavailability	-1.01%	Downtime depends on the diagnostic response time, stock of spare equipment and the repair response time. Further, loss in generation due to unavailability of plant and grid are also accounted for in downtime losses. It has considered 1% downtime on account of plant/grid unavailability and unused energy due to grid limitations is on

Parameters	Value considered	Our Remarks
		Higher side industry practise is between 0.5% - 1% only. Unavailability on higher side
AC Ohmic	-2.54%	This includes ohmic losses in the cable from the inverter until the 400kV take-off point. Based on review of the AC cable schedule provided by the Developer and the corresponding loss calculations, we found it on Higher side because of multiple AC station and Pooling arrangements. These losses are in the range of the industry practices (As per Pvsyst)

Energy Simulation Result is given below:

Table 29: Energy Simulation Assessment

Particulars	Details
Annual Degradation	0.42%
DC Capacity (MW)	840.00
Net Energy Generation (P50) In GWh	1584.00
Net Energy Generation (P75) In GWh	1537.80
Net Energy Generation (P90) In GWh	1496.10
Sum (P50) In GWh	21.52%
Sum (P75) In GWh	20.90%
Sum (P90) In GWh	20.33%
Net PLF (P90)	**20.33%**

Based on the above Energy Simulation Assessment, the projected generation for 25 years is given below:

Table 30: Projected Generation from Solar Plant

Year	P50			P75			P90		
	Generation (MWh)	DCPLF	ACPLF	Generation (MWh)	DCPLF	ACPLF	Generation (MWh)	DCPLF	ACPLF
1	1584000	21.53%	30.14%	1537800	20.90%	29.26%	1496100	20.33%	28.46%
2	1577347	21.44%	30.01%	1531341	20.81%	29.14%	1489816	20.25%	28.35%
3	1570722	21.35%	29.88%	1524910	20.72%	29.01%	1483559	20.16%	28.23%
4	1564125	21.26%	29.76%	1518505	20.64%	28.89%	1477328	20.08%	28.11%
5	1557556	21.17%	29.63%	1512127	20.55%	28.77%	1471123	19.99%	27.99%
6	1551014	21.08%	29.51%	1505776	20.46%	28.65%	1464945	19.91%	27.87%